

fields of the project, and we can leave the other fields to take their default options. Then click on 'Create' at the bottom of the tab as shown in Figure 7.

2. Adding client nodes

A 'Node Sources' tab, as shown in Figure 8, is opened as soon as we click on 'Create' in Step 1.

The path of the *resources.xml* file of this project can be seen. Click on the 'Edit' button and select the 'Writeable' option and save it. Now we will be able to edit

that file. The *resources.xml* file contains the node definitions. The Rundeck server reads this file and recognises the nodes. By default, the localhost is added as a node and the default *resources.xml* is shown in Figure 9.

Here, let us take a Docker container and add it as a node. Add lines to the *resources.xml* file as shown in Figure 10, and save it. 172.17.0.2 is the IP address of the Docker container and it is given as the hostname. The hostname

has to be unique across all the nodes. The values of 'username', 'osName' and 'osVersion' can vary depending on the type of the node.

On the Web UI of Rundeck, we can see that the client node has been added in the 'Nodes' tab.

3. Setting up SSH keys to authenticate the client nodes

For Rundeck to deploy applications or to execute jobs on the nodes, the Rundeck server has to log in into the nodes remotely. Now comes the question of security — how can we ensure that the system that is logging into the nodes is actually the server and not an attacker trying to gain unauthorised access to the node? The client nodes should be able to provide access only to the server and not to any other system. For this, we use SSH to authenticate the server, and allow it to execute commands in the nodes. SSH stands for Secure Shell, and it uses the public-private key encryption method for secure remote authentication.

Create a *.ssh* folder in the */var/lib/rundeck* directory of the host system and generate SSH keys in the '*id_rsa*' file. The private key is stored in the file '*id_rsa*' and the public key is stored in the file '*id_rsa.pub*'. In the *.ssh* folder that we have created, run the following command to generate the keys:

```
$ sudo ssh-keygen
```

When asked for the name of the file to save the keys, type '*id_rsa*'. We can use any other file name but usually '*id_rsa*' is preferred for SSH keys. Here, we are leaving the passphrase empty, but it can be given any value of your choice.

We have to copy the data in the '*id_rsa.pub*' file (which contains the SSH public key) and add it in the node.

Execute the following commands in the root directory of the node to add the public key of the host system:

```
$ mkdir .ssh
$ cd ~/.ssh
$ cat >> authorized_keys [6]
```

```
<?xml version="1.0" encoding="UTF-8"?>
<project>
  <node name="localhost" description="Rundeck server node" tags="" hostname="localhost" osArch="amd64" osFamily="unix" osName="Linux" osVersion="4.4.0-116-generic" username="rundeck"/>
</project>
```

Figure 9: Default *resources.xml* file

```
<?xml version="1.0" encoding="UTF-8"?>
<project>
  <node name="localhost" description="Rundeck server node" tags="" hostname="localhost" osArch="amd64" osFamily="unix" osName="Linux" osVersion="4.4.0-116-generic" username="rundeck"/>
  <node name="client" description="Client Node" tags="" hostname="172.17.0.2" osArch="amd64" osFamily="unix" osName="Linux" osVersion="4.13.0-36-generic" username="root" sudo-command-enabled="true" sudo-password-option="option.sudoPassword"/>
</project>
```

Figure 10: *Resources.xml* after adding a node

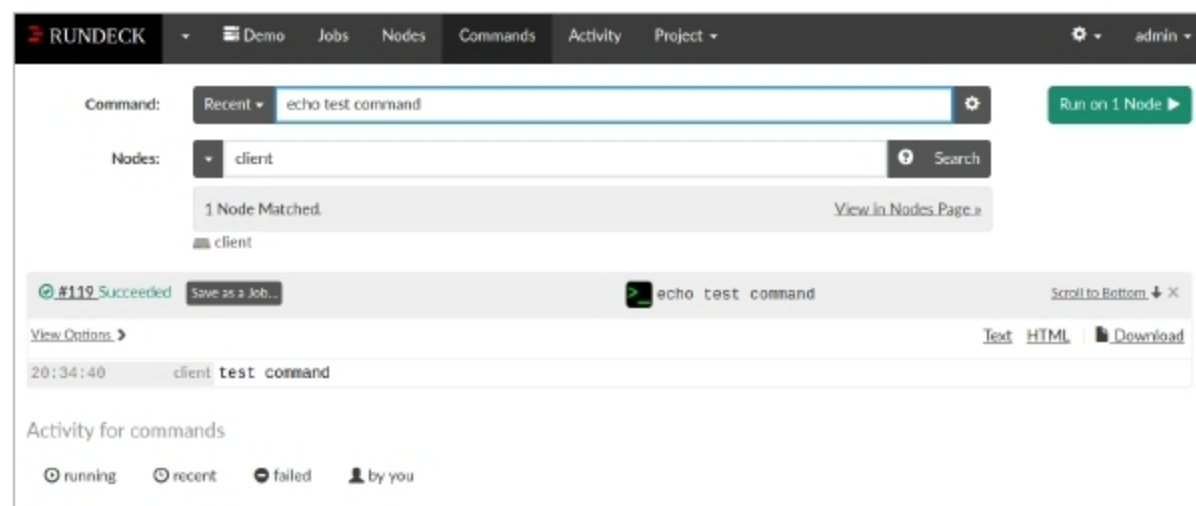


Figure 11: Test the command on the node

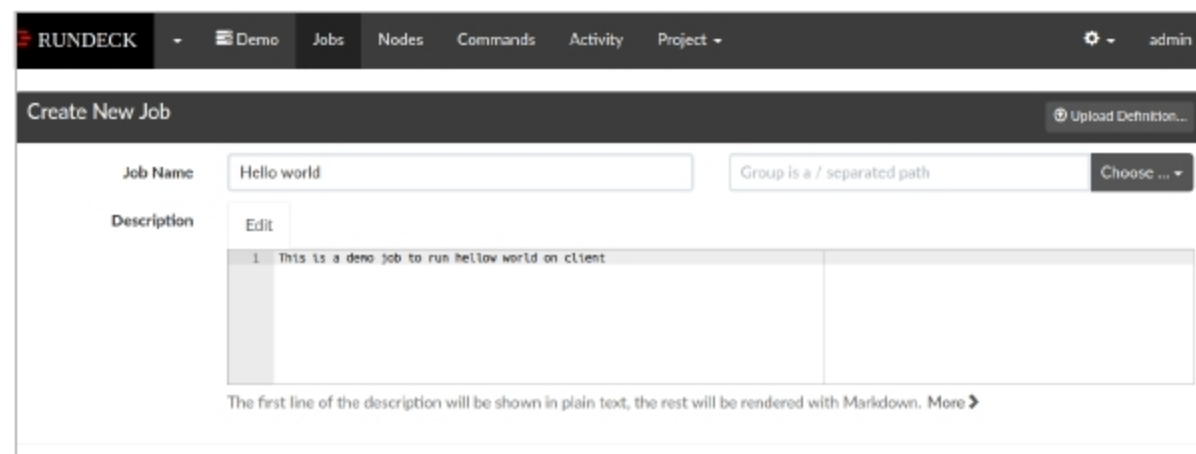


Figure 12: Creating a new job